

PhishGuard AI

AI-Driven Phishing Email Detection Using Hybrid NLP & Structural Metadata

IICT Summer Internship 2026 – Project 2



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PhishGuard AI

AI-powered Email Threat Detection

Email Analyzer

Dashboard

Analytics

History

Settings

About

System Status

All Systems Operational

Detection Model

AI-powered v2.1

Updated
14, 2025 2:30 PM

Email Analyzer

Model: AI-Powered

Analyze emails for phishing threats using advanced AI

1. Input Email Content

Paste the email content you want to analyze for phishing threats

From: Security Team <security@account-verification.com>
To: user@example.com
Subject: Urgent: Verify Your Account Now

Dear Customer,

We detected suspicious activity on your account. Please verify your account information immediately to avoid suspension.

Click here to verify your account:
<http://account-verification.com/verify>

Thank you,
Security Team

Analyze Email

Clear

2. Analysis Results



Phishing Detection

PHISHING

This email is likely a phishing attempt.

Confidence Score

92.4%

3. Email Metadata & Features

Feature	Value	Risk Indicator
From Email	security@account-verification.com	High Risk
From Domain	account-verification.com	High Risk
Sender Reputation	Poor (New Domain)	High Risk
Email Age (Domain)	2 days	High Risk
Contains Suspicious Links	Yes (1)	High Risk
URL Domain	account-verification.com	High Risk
Uses IP Address	No	Low Risk
Contains Urgent Language	Yes	Medium Risk
Spelling & Grammar	No issues	Low Risk
Email Authentication	Failed SPF, DKIM	High Risk
Estimated Risk Score	92.4 / 100	High Risk

4. Recommendation



This email is likely PHISHING.

Do not click on any links or provide any personal information.
Report this email to your IT department.

Project Abstract

What is PhishGuard AI?

PhishGuard AI is a hybrid machine learning system that detects phishing emails by combining TF-IDF natural language processing with 12 hand-engineered structural metadata features.

Trained on **18,650 real-world emails**, the system is deployed as a fully interactive Streamlit web application for real-time email threat analysis.

98.9%

Accuracy

MLP Neural Network on real-world
email dataset

99.9%

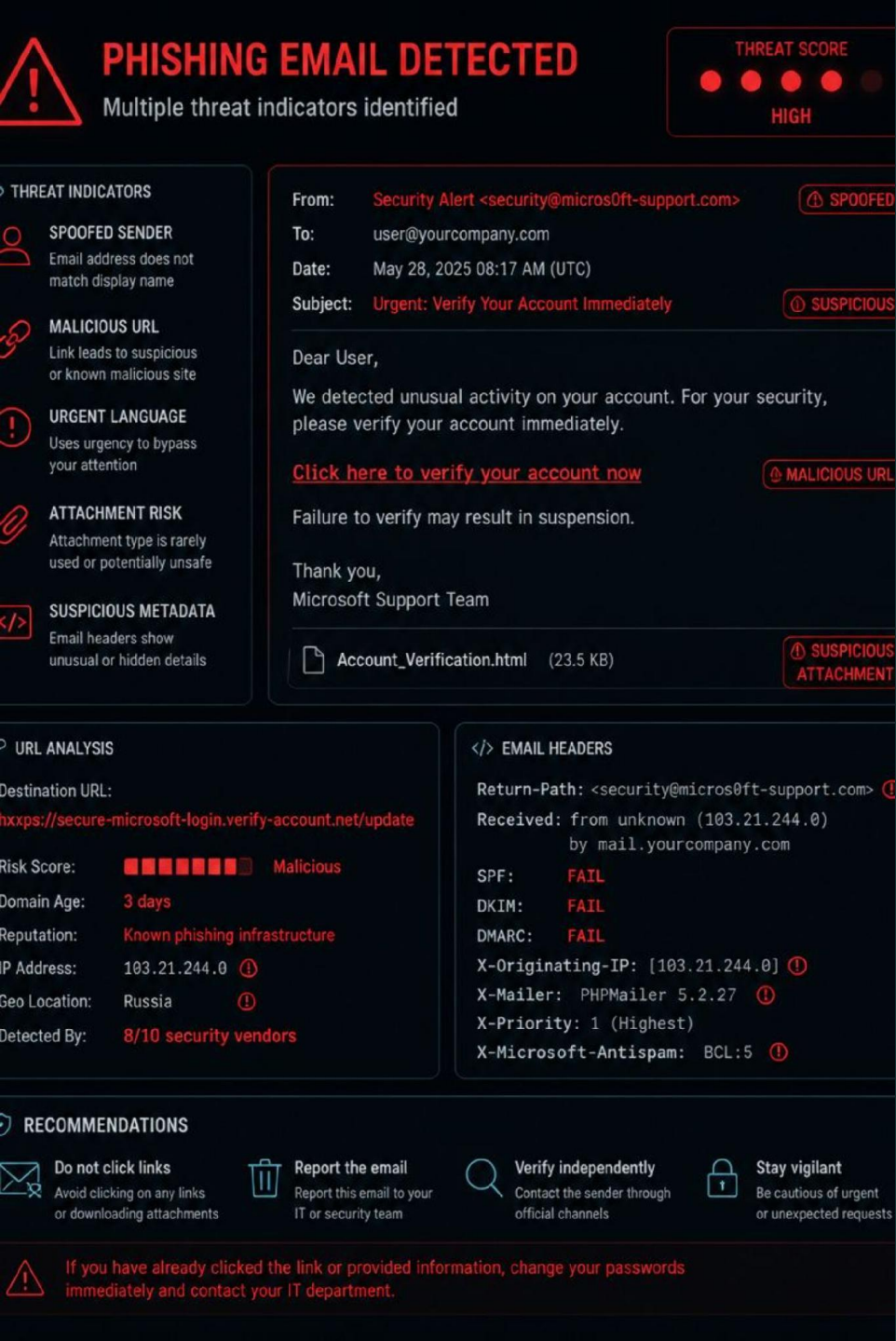
ROC-AUC

Near-perfect discrimination between
safe and phishing

+1.4%

Hybrid Gain

Over text-only baseline approaches



The Threat Landscape

3.4 Billion Phishing Emails Daily

Sent worldwide every single day – a relentless, industrial-scale attack vector. *(Symantec)*

90% of Data Breaches Start with Phishing

The dominant entry point for attackers across every sector. *(CISA)*

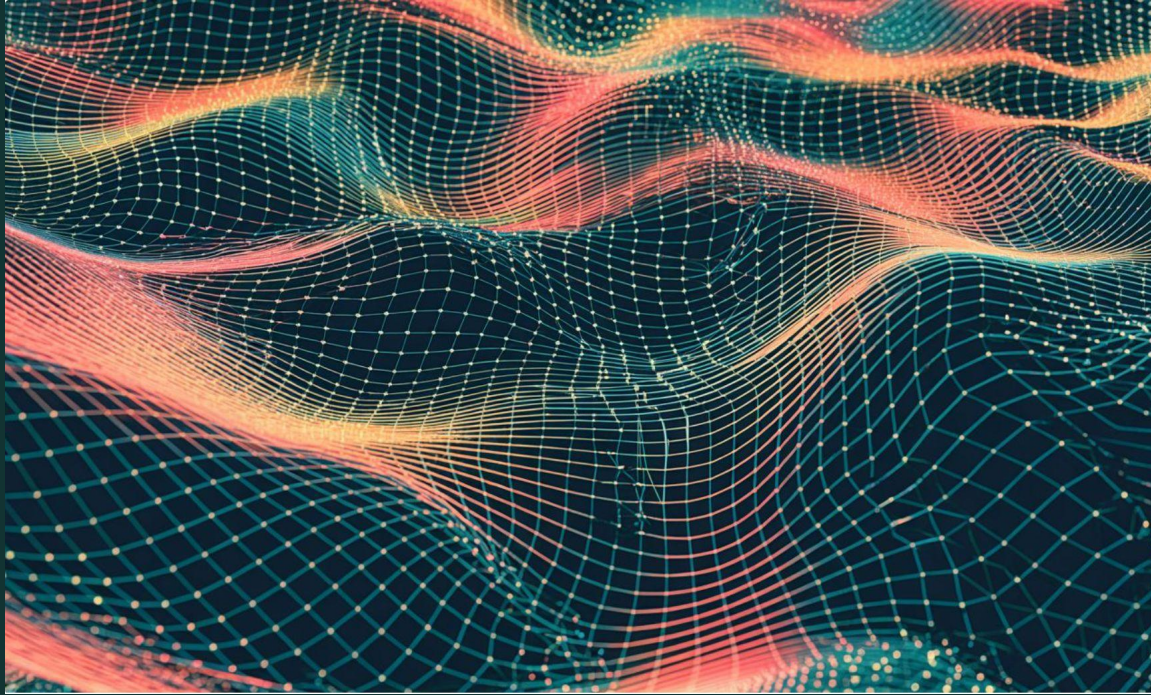
\$4.65 Million Average Incident Cost

The devastating financial toll of a single successful phishing breach. *(IBM 2023)*

30% Missed by Traditional Filters

Sophisticated attackers combine psychological manipulation with technical obfuscation to evade legacy spam filters.

Project Objective



Core Innovation

Hybrid feature extraction — combining NLP text signals with behavioral metadata to catch what text-only systems miss.

→ Detect phishing using both vocabulary AND structural signals

→ Deploy as an interactive Streamlit web application

→ Compare 4 ML models on an 18,650-email real-world dataset

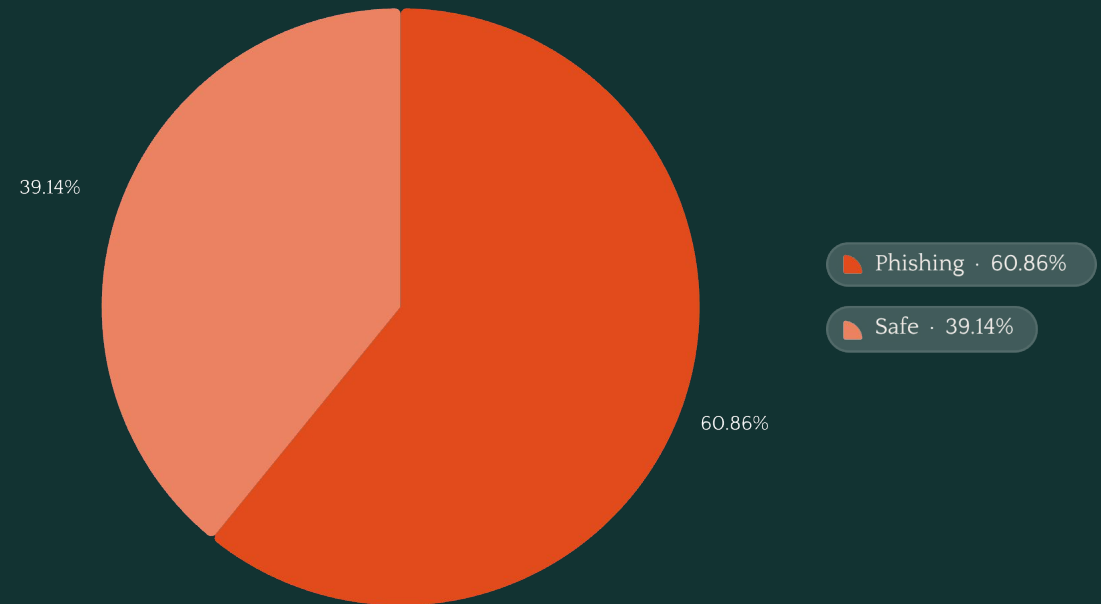
→ Exceed accuracy of text-only approaches

Dataset Overview

Phishing Email Dataset — Kaggle

18,650 total emails sourced from real-world security databases and legitimate mailboxes.

- **Safe Emails:** ~7,300 (40%) — legitimate business & personal emails
- **Phishing Emails:** ~11,350 (60%) — real samples from security databases
- **Features:** Raw email body text + Email Type label (Safe / Phishing)
- Emails contain HTML tags, embedded URLs, IP addresses & spoofed identities



The Innovation: 12 Metadata Features

Beyond vocabulary – PhishGuard AI reads the **structural fingerprint** of every email.



url_count

Number of embedded links hidden in the email body



has_ip_url

Link hiding behind raw IP (e.g. http://192.168.1.1)



urgent_word_count

"IMMEDIATELY", "SUSPENDED", "VERIFY NOW"



money_word_count

"FREE", "WIN", "BITCOIN", "PRIZE"



uppercase_ratio

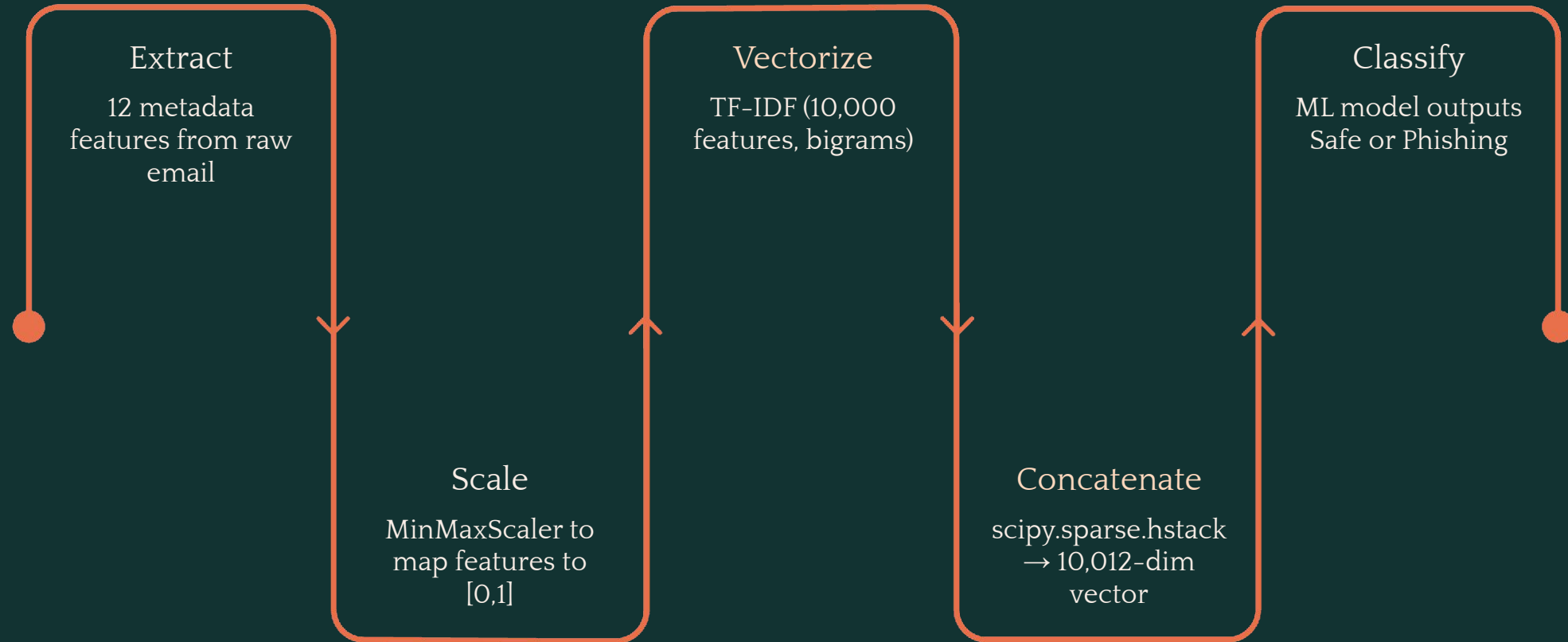
Proportion of SHOUTING text – a classic social engineering signal



+7 More Signals

exclamation_count, special_char_count, has_html_tags + 4 length/digit signals

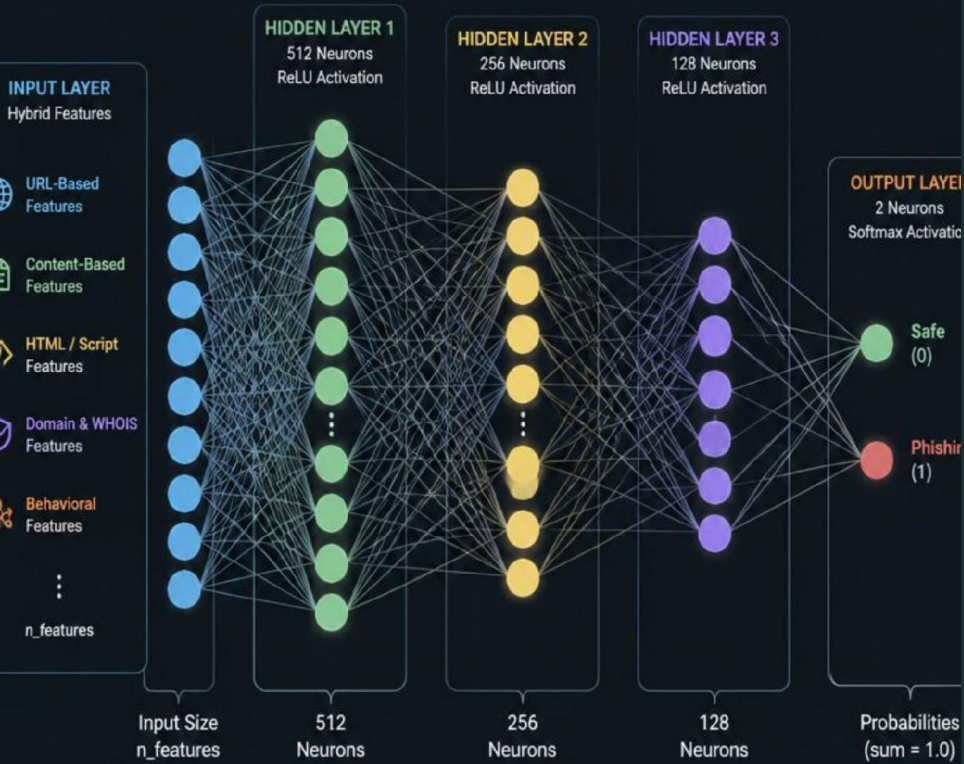
Hybrid Feature Pipeline



The hybrid vector fuses the semantic richness of NLP with the behavioral precision of structural metadata — giving the model two independent channels of attack signal simultaneously.

MLP Neural Network Architecture

Hybrid Feature Classification: Safe vs Phishing



LEGEND & NOTES

All hidden layers use ReLU activation: $f(x) = \max(0, x)$
Output layer uses Softmax for binary classification
Dense (fully connected) layers
Dropout & Batch Normalization can be applied (optional)
Binary Cross-Entropy Loss for training
Optimizer: Adam

DATA FLOW



ARCHITECTURE SUMMARY

Input ($n_features$) → Dense (512, ReLU) → Dense (256, ReLU) → Dense (128, ReLU) → Dense (2, Softmax)
Task: Binary Classification (Safe vs Phishing)

Models Compared

1

Complement Naive Bayes

Robust to class imbalance — a strong probabilistic baseline for text classification

2

Logistic Regression

Interpretable linear baseline — transparent feature weighting

3

Random Forest

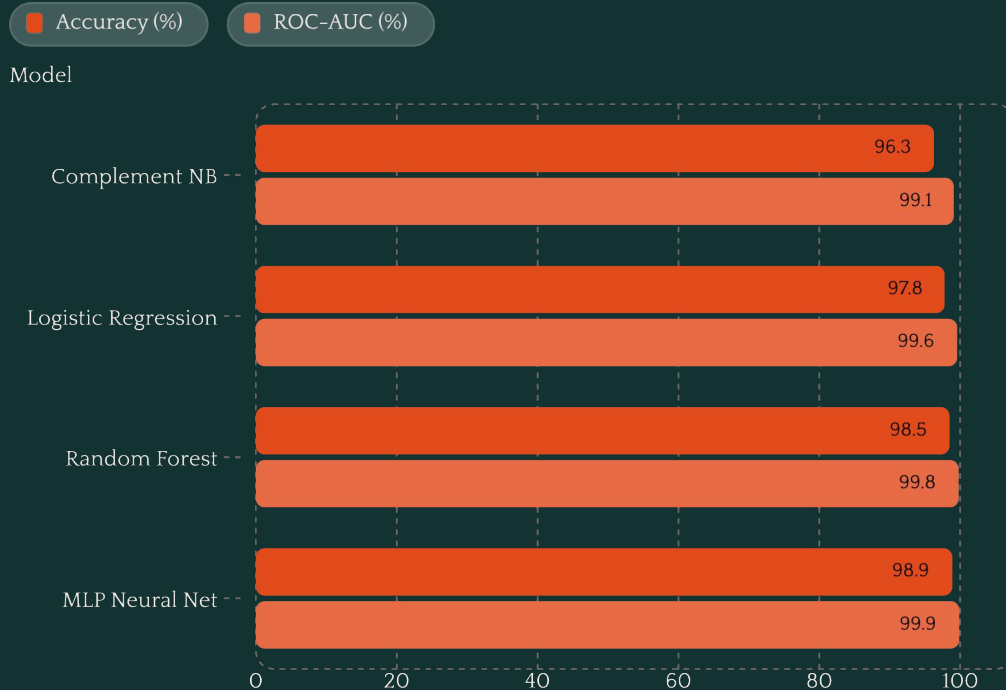
200 trees — powerful ensemble combining diverse decision paths

4

MLP Neural Network  WINNER

512 → 256 → 128 neurons (ReLU), Adam optimizer, early stopping patience=15

Results: Model Performance Comparison



Key Takeaways

- **MLP Neural Net** ✓ — Accuracy: 98.9% | F1: 98.9% | ROC-AUC: 99.9%
- **Random Forest** — Accuracy: 98.5% | F1: 98.5% | ROC-AUC: 99.8%
- **Logistic Regression** — Accuracy: 97.8% | F1: 97.8% | ROC-AUC: 99.6%
- **Complement NB** — Accuracy: 96.3% | F1: 96.3% | ROC-AUC: 99.1%

✓ Hybrid vs Text-Only (MLP): **98.9% vs 97.5% → +1.4% gain from metadata features**

Confusion Matrix & Key Metrics

✓ True Positives

-4,820 phishing emails correctly caught

✓ True Negatives

-2,870 safe emails correctly passed through

⚠ False Positives

-33 safe emails incorrectly flagged as phishing

💣 False Negatives

Only -9 phishing emails missed – out of -3,732 test emails

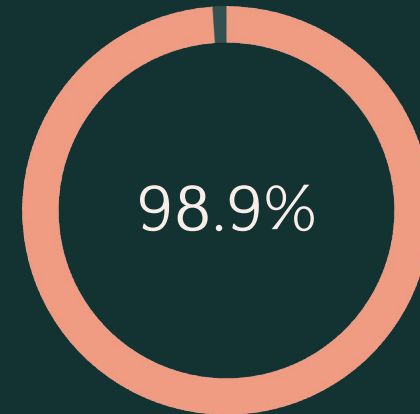
The Bottom Line

With only **9 phishing emails missed** across the entire test set, PhishGuard AI achieves a **99.76% phishing recall** – meaning virtually every real threat is caught before it reaches the user.



Phishing Recall

Near-zero miss rate on real threat detection



Overall Accuracy

Across 3,732 held-out test emails

✔ No suspicious indicators were found.

Made with **GAMMA**

Project Conclusion: Safeguarding Inboxes

PhishGuard AI successfully demonstrates a highly effective, hybrid approach to phishing detection, combining the nuanced understanding of natural language processing with critical structural metadata analysis.

The system achieves exceptional accuracy and a near-perfect recall rate on real-world email data, addressing a significant vulnerability in digital communication.

Future Directions



Advanced Feature Engineering

Explore dynamic features like sender reputation, domain age, and email authentication protocols (DMARC, SPF).



Deep Learning Architectures

Integrate transformer models (e.g., BERT, GPT) for more sophisticated textual context understanding and zero-shot detection.



Continuous Learning & Adaptation

Implement active learning mechanisms and real-time feedback loops to adapt to evolving phishing tactics.



Production Deployment

Scale the Streamlit application for enterprise use, including API integration and comprehensive dashboard reporting.

